

Module 3 — Read 2

Two categorical and two numerical variables

Module 3, Read 2: Describing Pairs of Variables

Complete Read 1 and Quiz A first. This reading finishes the descriptive section of our toolbox: two categorical variables (two-way tables and stacked bars), then two numerical variables (scatterplots).

Learning outcomes

Two sections — use Check yourself before [Quiz B](#).

After this reading you should be able to:

Section 4: Two categorical variables

Two categorical — contingency tables, proportions, stacked bar charts

- Build a two-way table of counts and compute joint, row, and column proportions (know which denominator you use).
- Draw stacked bar charts for counts and 100% stacked bars (`position = "fill"`) to show proportions within bars.
- Implement tables and plots with `count()`, `group_by()`, `mutate()`, and arrange with `pivot_wider()`.

Section 5: Two numerical variables

Two numerical — scatterplots, describing relationships, Pearson r

- Make and interpret scatterplots for two numerical variables (form, direction, strength).
- Compute and interpret Pearson correlation r as a unitless measure of linear association ($-1 \leq r \leq 1$).
- Recognize when r is misleading (nonlinear form, outliers).

Section 4: Two categorical variables

i Bridge from Section 3

Section 3 crossed one numerical with one categorical. Here both variables are categorical, like if you were asking two questions, each of which has non-numerical answers. We summarize two categorical variables with a two-way table of counts, then joint, row, and column proportions, and stacked bar charts (counts or 100% fill).

i Supplemental videos: two-way tables and 100% stacked bars

Resource	Type
Two-way relative frequency tables — Khan Academy	Video
Conditional relative frequencies and association — Khan Academy	Video
Stacked and proportional bar charts in ggplot2 — Andrew Couch	Video

Why two categorical variables together

Each observation has a level of variable A (e.g. species) and a level of variable B (e.g. habitat). We summarize how the combinations co-occur: counts in each cell of a table, then graphs can show these counts or proportions at a glance.

Small example by hand (two binary variables)

Setting. Birds counted in two habitats: Urban or Rural. Species recorded as Robin or Sparrow (ignore other species here).

Counts:

	Urban	Rural	Row total
Robin	8	2	10
Sparrow	4	6	10
Column total	12	8	20 = grand total N

So there are 20 birds. The joint count in each cell is a count of birds with that (species, habitat) pair.

Joint proportions (of the grand total)

Each cell count divided by $N = 20$ answers: *What fraction of all birds fall in this cell?*

- Robin & Urban: $8/20 = 0.40$
- Robin & Rural: $2/20 = 0.10$
- Sparrow & Urban: $4/20 = 0.20$
- Sparrow & Rural: $6/20 = 0.30$

Check: $0.40 + 0.10 + 0.20 + 0.30 = 1$.

Row proportions (conditional on species)

Within each species, divide by that row's total (10). Answers: *Among Robins only, what fraction were Urban vs Rural?* (and similarly for Sparrows.)

Robin: Urban $8/10 = 0.80$, Rural $2/10 = 0.20$.

Sparrow: Urban $4/10 = 0.40$, Rural $6/10 = 0.60$.

Each row sums to 1.

Column proportions (conditional on habitat)

Within each habitat, divide by that column's total. Answers: *Among Urban birds, what fraction were Robin vs Sparrow?*

Urban: Robin $8/12 \approx 0.67$, Sparrow $4/12 \approx 0.33$ (column total 12).

Rural: Robin $2/8 = 0.25$, Sparrow $6/8 = 0.75$ (column total 8).

Each column sums to 1.

Note: The same four counts support three different stories depending on whether you normalize by everything (joint or "grand"), by rows (species), or by columns (habitat). Be explicit about which denominator you use when you write a sentence or label a plot.

Example 1: The bird table in R's tidyverse

We store one row per bird: variables `species` and `habitat`.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

birds <- tibble(
  species = c(rep("Robin", 10), rep("Sparrow", 10)),
  habitat = c(rep(c("Urban", "Rural"), c(8, 2)), rep(c("Urban", "Rural"), c(4, 6)))
) |>
  mutate(
    species = factor(species, levels = c("Robin", "Sparrow")),
    habitat = factor(habitat, levels = c("Urban", "Rural"))
  )
```

Two-way table of counts

```
birds |>
  count(species, habitat) |>
  pivot_wider(names_from = habitat, values_from = n, values_fill = 0)
```

```
# A tibble: 2 x 3
  species Urban Rural
  <fct>   <int> <int>
1 Robin         8     2
2 Sparrow        4     6
```

Joint, row, and column proportions (long format)

```
joint <- birds |>
  count(species, habitat) |>
  mutate(prop_joint = n / sum(n))

row_prop <- birds |>
  count(species, habitat) |>
  group_by(species) |>
  mutate(prop_row = n / sum(n)) |>
  ungroup()

col_prop <- birds |>
  count(species, habitat) |>
  group_by(habitat) |>
  mutate(prop_col = n / sum(n)) |>
  ungroup()

joint
```

```
# A tibble: 4 x 4
```

	species	habitat	n	prop_joint
	<fct>	<fct>	<int>	<dbl>
1	Robin	Urban	8	0.4
2	Robin	Rural	2	0.1
3	Sparrow	Urban	4	0.2
4	Sparrow	Rural	6	0.3

```
row_prop
```

```
# A tibble: 4 x 4
  species habitat     n prop_row
  <fct>   <fct> <int>   <dbl>
1 Robin   Urban     8     0.8
2 Robin   Rural     2     0.2
3 Sparrow Urban     4     0.4
4 Sparrow Rural     6     0.6
```

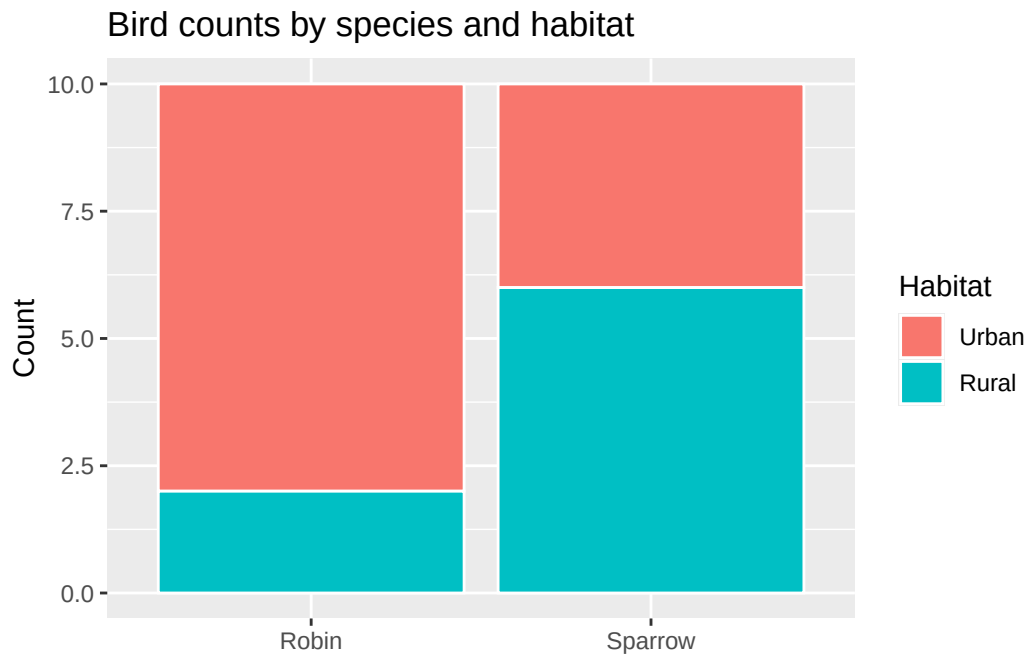
```
col_prop
```

```
# A tibble: 4 x 4
  species habitat     n prop_col
  <fct>   <fct> <int>   <dbl>
1 Robin   Urban     8    0.667
2 Robin   Rural     2    0.25
3 Sparrow Urban     4    0.333
4 Sparrow Rural     6    0.75
```

Stacked bar chart of counts

Bars show species; segments show habitat; height of each segment is the cell count.

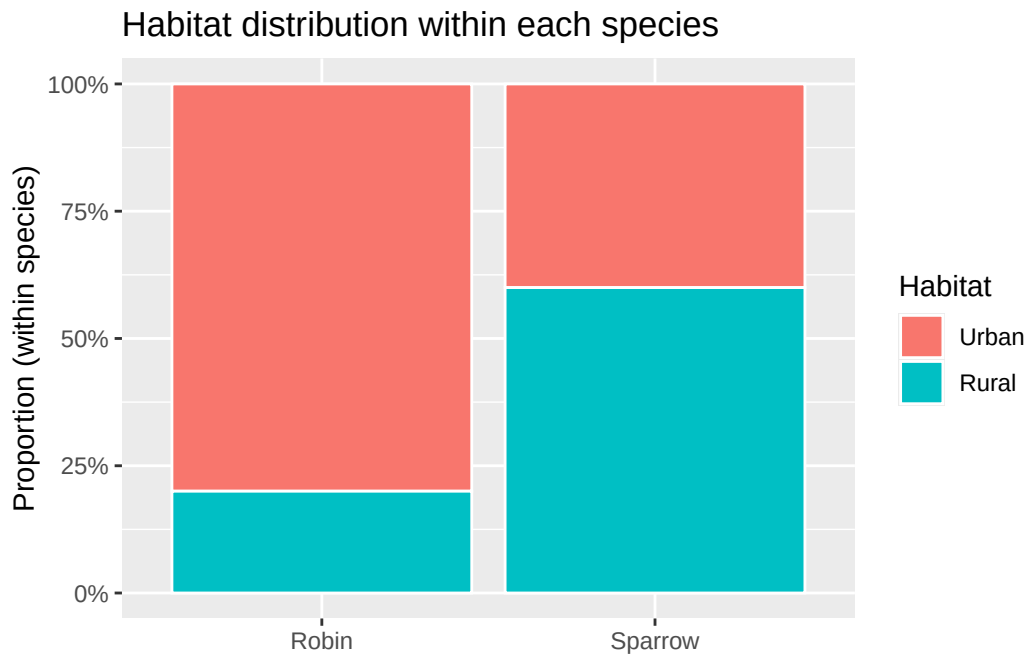
```
birds |>
  count(species, habitat) |>
  ggplot(aes(x = species, y = n, fill = habitat)) +
  geom_col(color = "white") +
  labs(
    title = "Bird counts by species and habitat",
    x = NULL,
    y = "Count",
    fill = "Habitat"
  )
```



Stacked bar chart with 100% within each species (row proportions)

`position = "fill"` stretches each bar to height 1; segment heights match row proportions (`prop_row`). Good for comparing habitat mix between species when raw counts differ.

```
birds |>
  count(species, habitat) |>
  ggplot(aes(x = species, y = n, fill = habitat)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
  labs(
    title = "Habitat distribution within each species",
    x = NULL,
    y = "Proportion (within species)",
    fill = "Habitat"
  )
```



Notice that you swap what is on the x-axis and what is `fill` if the research question reads more naturally the other way (e.g. habitat on the x-axis, species as fill).

Example 2: mpg: class vs drv (several categories each)

Each row is a car model. We cross vehicle class and drive type.

```
data("mpg", package = "ggplot2")
```

```
mpg |>
  count(class, drv) |>
  pivot_wider(names_from = drv, values_from = n, values_fill = 0)
```

```
# A tibble: 7 x 4
  class      r   `4`   f
  <chr>    <int> <int> <int>
1 2seater     5     0     0
2 compact     0    12    35
3 midsize     0     3    38
4 minivan     0     0    11
5 pickup      0    33     0
6 subcompact  9     4    22
7 suv        11    51     0
```

Row proportions: among cars in a given class, what fraction have each drv?

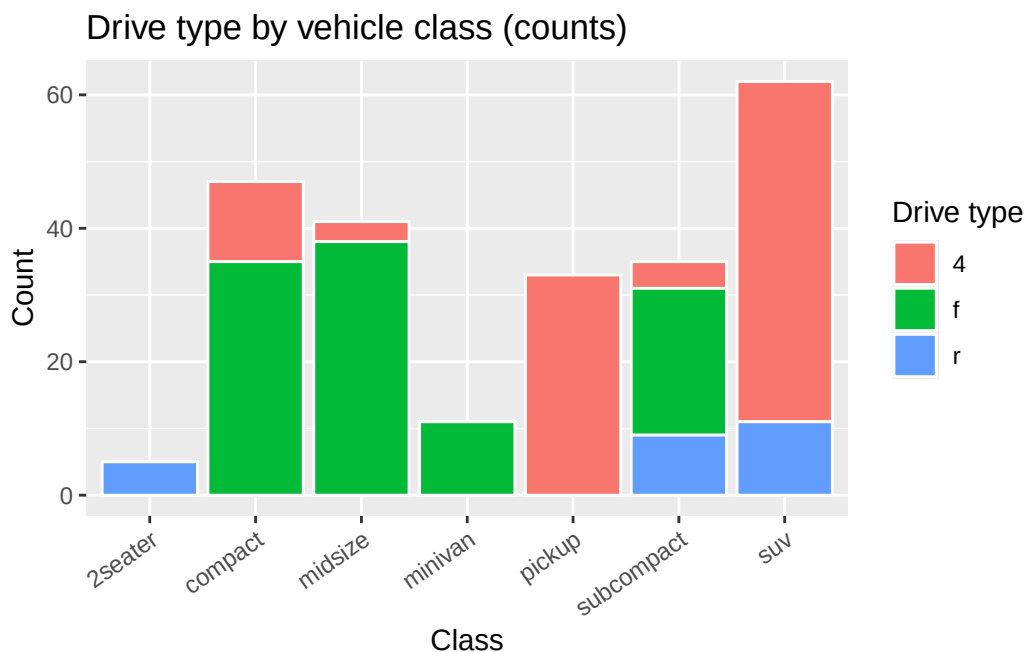
```
mpg |>
  count(class, drv) |>
  group_by(class) |>
  mutate(prop_within_class = n / sum(n)) |>
  ungroup() |>
  print(n = 20)
```

```
# A tibble: 12 x 4
  class     drv      n prop_within_class
  <chr>    <chr> <int>         <dbl>
1 2seater   r         5             1
2 compact  4        12          0.255
3 compact  f        35          0.745
4 midsize  4         3          0.0732
5 midsize  f        38          0.927
6 minivan  f        11             1
7 pickup   4        33             1
8 subcompact 4         4          0.114
9 subcompact f        22          0.629
10 subcompact r         9          0.257
11 suv      4        51          0.823
12 suv      r        11          0.177
```

Stacked bars (counts and 100% stacked):

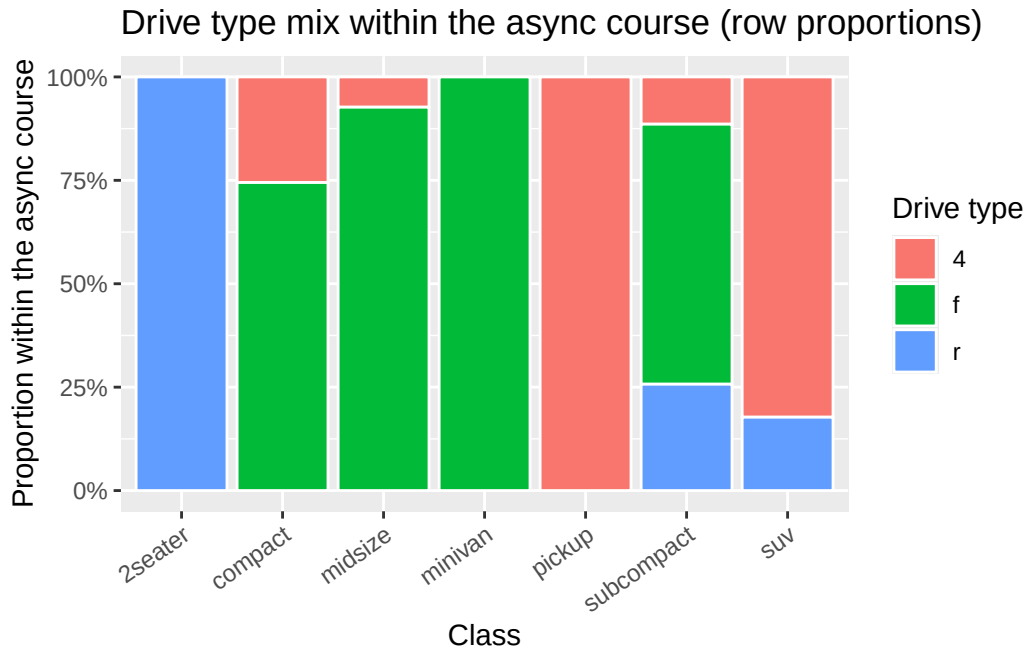
```
mpg_ct <- mpg |> count(class, drv)

ggplot(mpg_ct, aes(x = class, y = n, fill = drv)) +
  geom_col(color = "white") +
  labs(
    title = "Drive type by vehicle class (counts)",
    x = "Class",
    y = "Count",
    fill = "Drive type"
  ) +
  theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



```
ggplot(mpg_ct, aes(x = class, y = n, fill = drv)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
```

```
labs(
  title = "Drive type mix within the async course (row proportions)",
  x = "Class",
  y = "Proportion within the async course",
  fill = "Drive type"
) +
theme(axis.text.x = element_text(angle = 35, hjust = 1))
```



Example 3: TidyTuesday Palmer Penguins: species vs island

I load the published TidyTuesday file from GitHub with `read_csv()`. One could also download the CSV once and use a local path instead.

```
penguins <- read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-07-28/penguins.csv"
)

penguins |>
  count(species, island) |>
  pivot_wider(names_from = island, values_from = n, values_fill = 0)
```

```
# A tibble: 3 x 4
  species   Biscoe Dream Torgersen
  <chr>     <int> <int>     <int>
1 Adelie     44     56        52
2 Chinstrap    0     68         0
3 Gentoo    124     0         0
```

Column proportions: among birds on a given island, what fraction belong to each species?

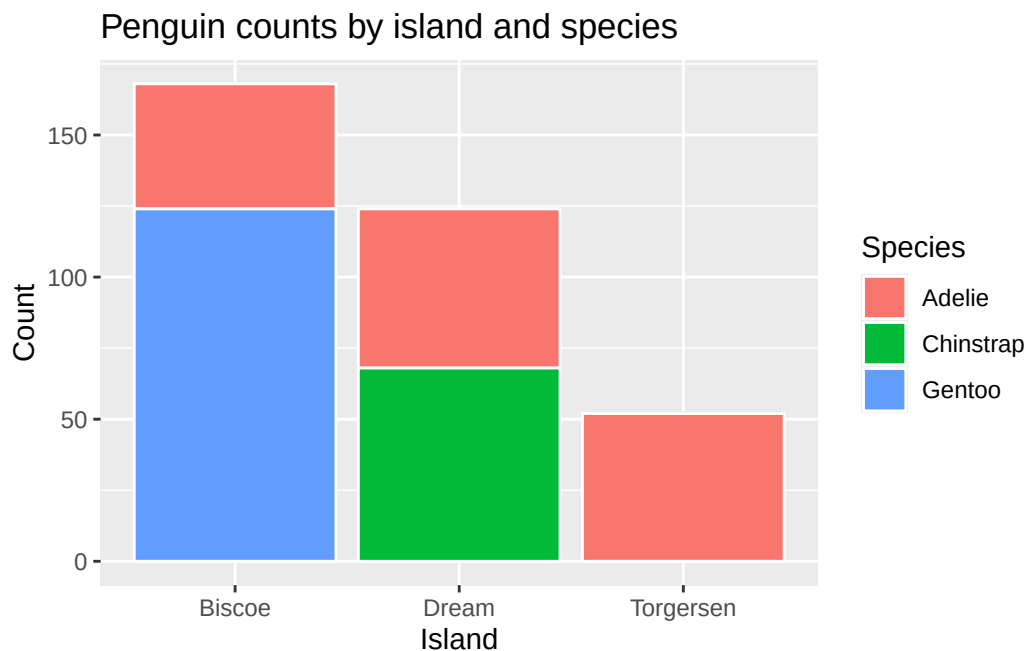
```
penguins |>
  drop_na(species, island) |>
  count(species, island) |>
  group_by(island) |>
  mutate(prop_within_island = n / sum(n)) |>
  ungroup()
```

```
# A tibble: 5 x 4
  species island      n prop_within_island
  <chr>   <chr>   <int>         <dbl>
1 Adelie  Biscoe      44         0.262
2 Adelie  Dream       56         0.452
3 Adelie  Torgersen   52          1
4 Chinstrap Dream     68         0.548
5 Gentoo  Biscoe     124         0.738
```

Stacked bars: counts and column-style emphasis: put island on the x-axis and species as fill so each bar is an island total and segments show species mix on that island.

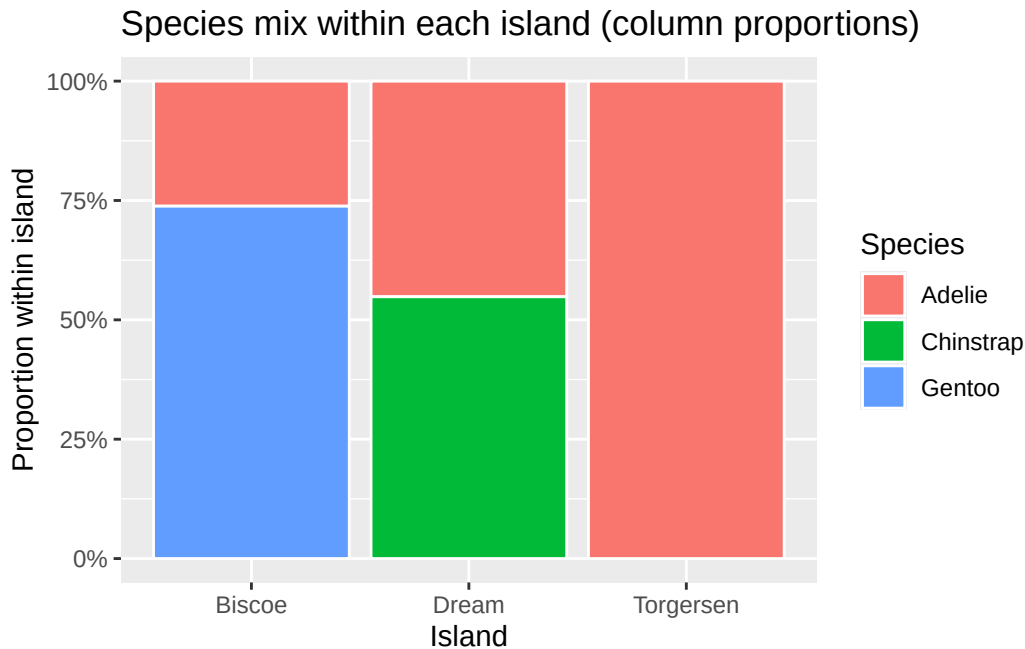
```
pg <- penguins |> drop_na(species, island) |> count(species, island)

ggplot(pg, aes(x = island, y = n, fill = species)) +
  geom_col(color = "white") +
  labs(
    title = "Penguin counts by island and species",
    x = "Island",
    y = "Count",
    fill = "Species"
  )
```



```
ggplot(pg, aes(x = island, y = n, fill = species)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::label_percent()) +
```

```
labs(
  title = "Species mix within each island (column proportions)",
  x = "Island",
  y = "Proportion within island",
  fill = "Species"
)
```



💡 Interpret the output — code comes in the demo lab

The chunks below use tidyverse to produce tables and figures. Focus on reading counts, proportions, summaries, and graphs. You will learn how to write these steps in [Demo 1](#) and [Demo 2](#).

Example 1 — Species × island (penguins)

```
library(tidyverse)
if (!requireNamespace("palmerpenguins", quietly = TRUE)) {
  install.packages("palmerpenguins", repos = "https://cloud.r-project.org")
}
library(palmerpenguins)

penguins |>
  count(species, island) |>
  pivot_wider(names_from = island, values_from = n, values_fill = 0)
```

```
# A tibble: 3 x 4
  species   Biscoe Dream Torgersen
  <chr>     <int> <int>   <int>
1 Adelie      44    56      52
2 Chinstrap    0    68       0
3 Gentoo    124     0       0
```

```
ggplot(penguins, aes(x = island, fill = species)) +
  geom_bar(color = "white") +
  labs(title = "Penguin counts by island", x = "Island", y = "Count", fill = "Species")
```

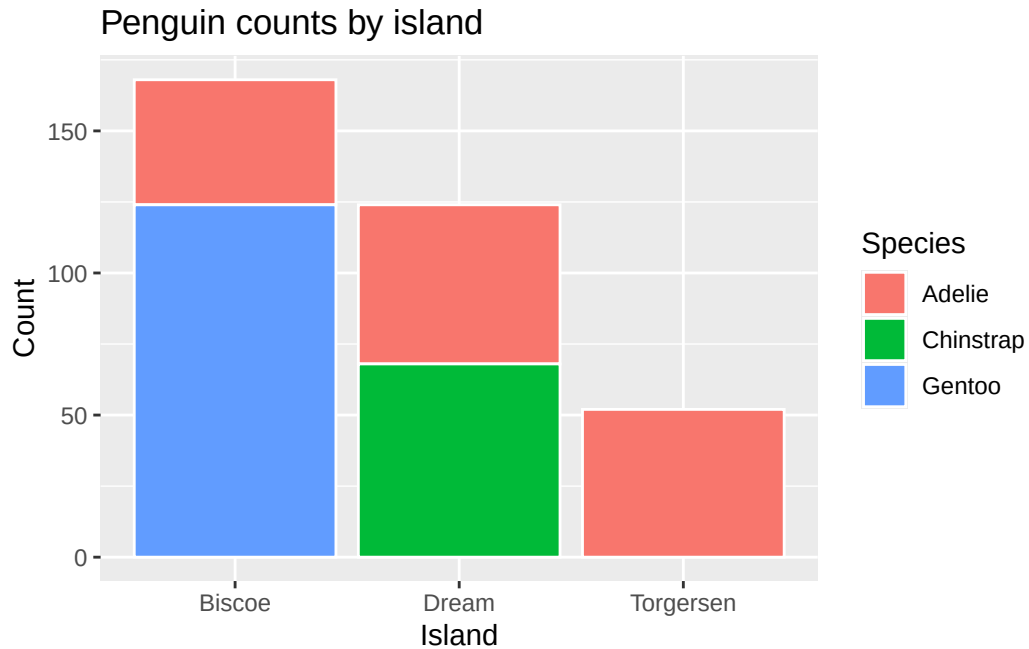


Figure 1: Stacked counts: species composition by island.

```
ggplot(penguins, aes(x = island, fill = species)) +
  geom_bar(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Species mix within island", x = "Island", y = "Proportion", fill = "Species")
```

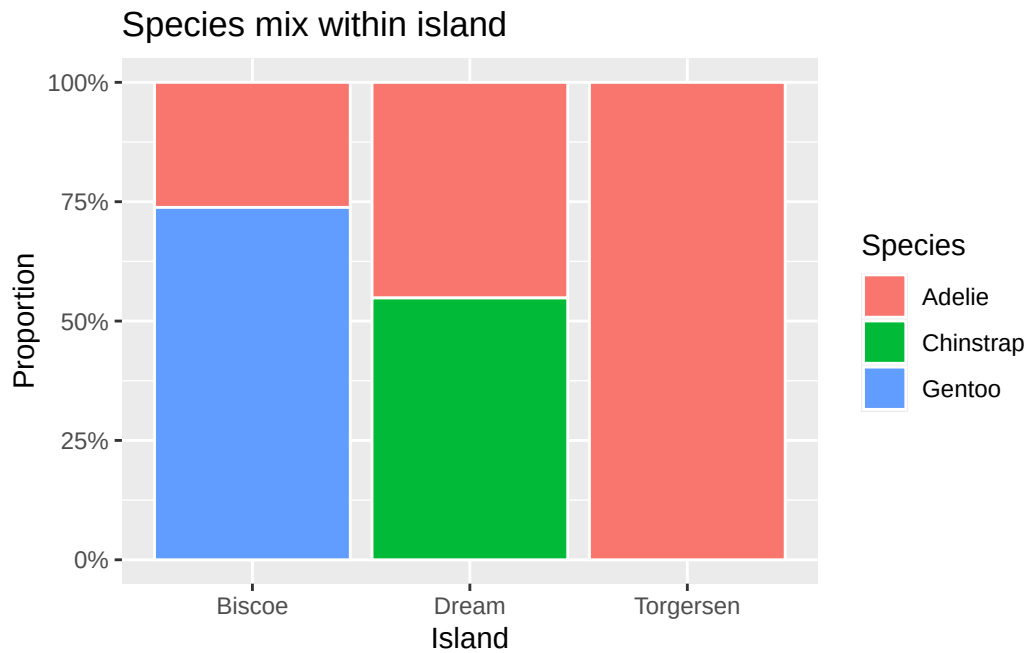


Figure 2: 100% stacked — species mix within each island.

Interpret: State one joint proportion, one row proportion, and one column proportion from this table.

Answer (one of many): joint: Gentoo & Biscoe = $124/344 \approx 0.36$; row: among Adelie, Torgersen = $52/152 \approx 0.34$; column: among Biscoe penguins, Gentoo = $124/168 \approx 0.74$.

Example 2 — Vehicle class × drive type (mpg)

```
data("mpg", package = "ggplot2")
mpg |> count(class, drv) |> print(n = 20)
```

```
# A tibble: 12 x 3
  class     drv     n
  <chr>    <chr> <int>
1 2seater   r         5
2 compact  4        12
3 compact  f        35
4 midsize  4         3
5 midsize  f        38
6 minivan  f        11
7 pickup   4        33
8 subcompact 4         4
9 subcompact f        22
10 subcompact r         9
11 suv      4        51
12 suv      r        11
```

```
ggplot(mpg, aes(x = class, fill = drv)) +
  geom_bar(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::percent) +
```

```
labs(title = "Drive type within class", x = "Class", y = "Proportion", fill = "Drive") +
theme(axis.text.x = element_text(angle = 35, hjust = 1))
```

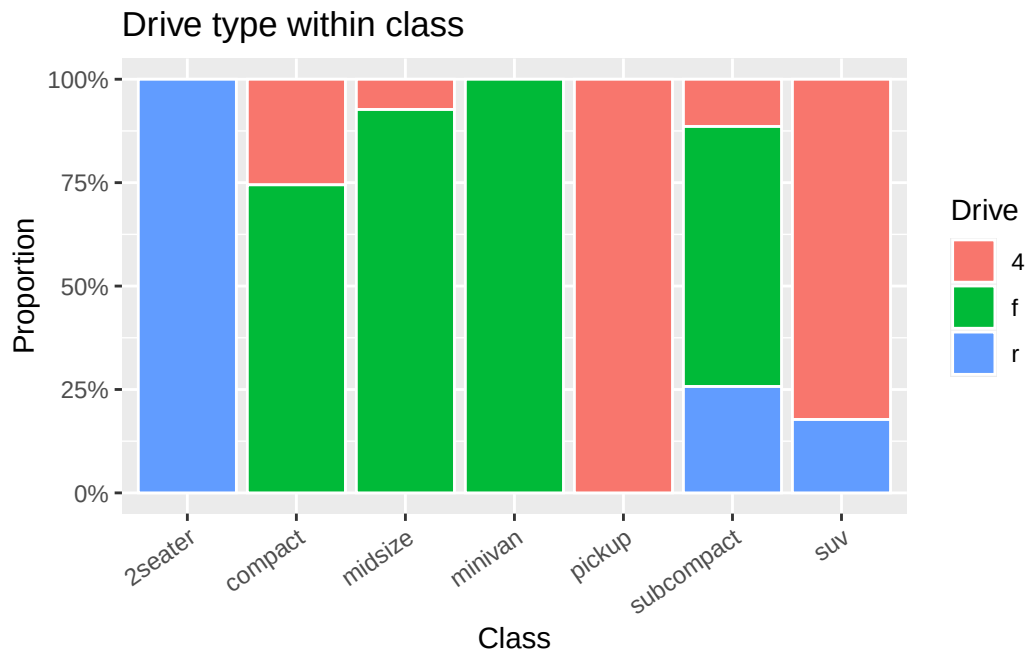


Figure 3: Drive-type mix within each vehicle class.

Interpret: Among suv models, which drive type is most common?

Answer: Four-wheel drive (4) is most common among suv models (the rest are rear-wheel, r; none are front-wheel).

Example 3 — Bird species × habitat (toy 2×2)

```
birds <- tibble(
  species = c(rep("Robin", 10), rep("Sparrow", 10)),
  habitat = c(rep(c("Urban", "Rural"), c(8, 2)), rep(c("Urban", "Rural"), c(4, 6)))
)
birds |> count(species, habitat) |> mutate(joint_prop = n / sum(n))
```

```
# A tibble: 4 x 4
  species habitat     n joint_prop
  <chr>   <chr>   <int>     <dbl>
1 Robin   Rural     2       0.1
2 Robin   Urban     8       0.4
3 Sparrow Rural     6       0.3
4 Sparrow Urban     4       0.2
```

```
birds |>
  count(species, habitat) |>
  ggplot(aes(x = species, y = n, fill = habitat)) +
  geom_col(position = "fill", color = "white") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Habitat within species", x = "Species", y = "Proportion", fill = "Habitat")
```

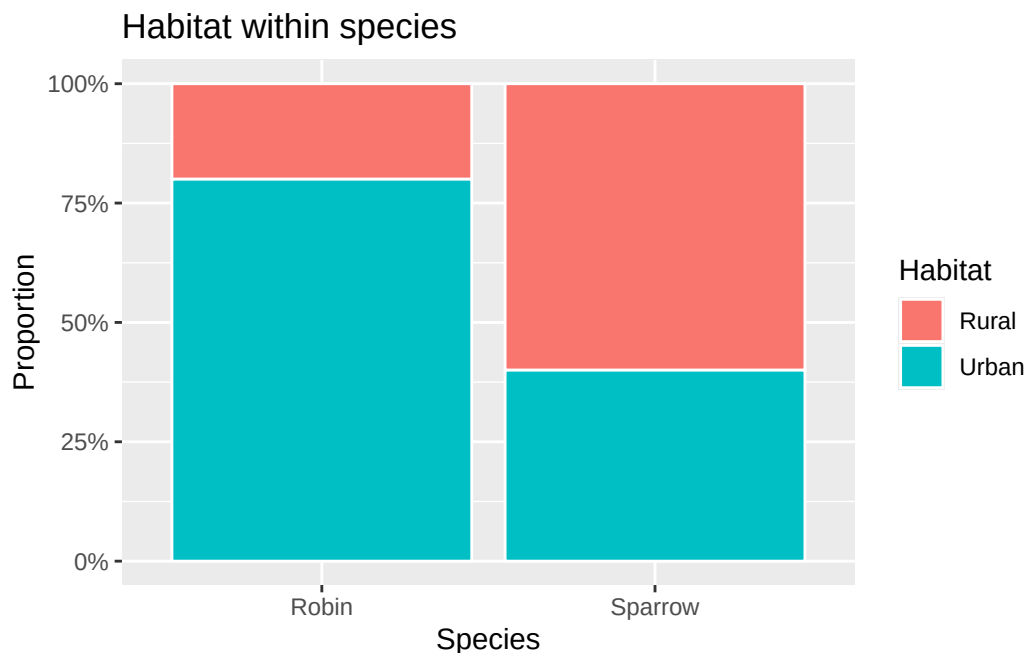


Figure 4: 100% stacked — habitat mix within each species (row proportions).

Interpret: What fraction of Robins are Urban?

Answer: $8/10 = 0.80$ (80%) of Robins are Urban.

i Compute by hand (small datasets)

Work these before checking R. Same numbers appear in [Quiz A](#) and [Synthesis](#) practice banks.

Hand dataset 1 — Species $\times$ habitat ($n = 20$)

	Urban	Rural	Row total
Robin	8	2	10
Sparrow	4	6	10
Col total	12	8	20

- Row proportion Sparrow Rural: $6/10 = 0.60$.
- Column proportion Urban Robin: $8/12 \approx 0.667$.
- Joint proportion Robin Urban: $8/20 = 0.40$.

Hand dataset 2 — Treatment $\times$ outcome ($n = 50$)

	Success	Failure	Row total
A	18	7	25
B	10	15	25
Col total	28	22	50

- Row proportion B Success: $10/25 = 0.40$.

- Column proportion Success that is A: $18/28 \approx 0.643$.
- Joint proportion B Failure: $15/50 = 0.30$.

Hand dataset 3 — Island $\times$ species (small)

	Biscoe	Dream	Row total
Adelie	6	4	10
Gentoo	9	1	10
Col total	15	5	20

- Row proportion Gentoo on Biscoe: $9/10 = 0.90$.
- Column proportion Dream that is Adelie: $4/5 = 0.80$.
- Joint proportion Gentoo Biscoe: $9/20 = 0.45$.

i From this sample to a population — inference reference note

Described here: counts and row/column proportions in a two-way table. **Population parameter:** for a 2×2 table, the difference in population proportions $p_1 - p_2$; more generally, whether the two categorical variables are **associated** (not independent) in the population.

Two groups, one proportion each (2×2): two-proportion test. If you want to **compare two population proportions** ($H_0 : p_1 = p_2$) or estimate their difference, then — after checking conditions (independent observations; expected cell counts large enough, about ≥ 5) — use the two-proportion test, which returns a confidence interval for $p_1 - p_2$ and a p -value.

Any size table: chi-square test of independence. More generally, test whether the two categorical variables are associated,

$$H_0 : \text{the two variables are independent} \quad \text{vs.} \quad H_a : \text{they are associated,}$$

using the chi-square statistic, which compares the **observed** cell counts to the counts you'd **expect** if the variables were independent,

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}, \quad E = \frac{(\text{row total})(\text{column total})}{N}.$$

If the variables are truly independent, O and E are close and χ^2 is small; a large χ^2 (small p -value) is evidence of association. The reference distribution has $\text{df} = (r - 1)(c - 1)$ for an $r \times c$ table. For a 2×2 table the chi-square test and the two-proportion test answer the same question. Let software do the work:

```
prop.test(c(x1, x2), c(n1, n2))    # 2x2: CI for p1 - p2 and test
chisq.test(table(var1, var2))      # general test of independence (any size table)
```

Both are introduced in **Module 4** (two-proportion and chi-square tests).

Check yourself — Two categorical variables

1. Birds: 8 Robin+Urban, 2 Robin+Rural, 4 Sparrow+Urban, 6 Sparrow+Rural ($N = 20$). Joint proportion Robin+Urban?
2. Same table: row proportion of Robins that are Urban?
3. What does `position = "fill"` do in a stacked bar chart?
4. Which function pivots a count table from long to wide?

Answers — Two categorical variables

1. $8/20 = 0.40$.
2. $8/10 = 0.80$.
3. Normalizes each bar to 100% so the plot shows proportions within each bar.
4. `pivot_wider()`.

Section 5: Two numerical variables

i Bridge from Section 4

Section 4 crossed two categorical variables. When both variables are numerical, plot a scatterplot, describe form, direction, and strength, and when form is roughly linear summarize with Pearson correlation r . (Fitting regression lines and prediction come in Module 5.)

i Supplemental videos — scatterplots

Resource	Type
Scatterplots: construction and reading — Khan Academy	Video
Describing trends in scatterplots — Khan Academy	Video

Why two numerical variables together

When both variables are numerical, we often ask how one changes with the other: does taller vegetation go with wetter soil? Does leaf nitrogen track soil nitrate? A scatterplot puts one variable on the horizontal axis and one on the vertical axis; each point is one observational unit (a tree, a nest, a transect).

Describing a scatterplot in words

Read the cloud of points as a whole.

Form (shape). Is the trend roughly a straight line (linear), bent (curved / nonlinear), split into clusters, or no clear pattern?

Direction. As x increases, does y tend to increase (positive association), decrease (negative association), or show no consistent up-or-down trend?

Strength. How tightly do points follow the trend in the form? Strong: points closely hug a line or curve. Moderate: signal is visible but some random scatter above and below is noticeable. Weak: only a faint drift. None: no useful trend (uniform random scatter around a horizontal band).

These labels are judgment calls in practice; the point is to communicate what you see before moving to formal models.

Examples (biology and field settings)

The plots below use made-up data for teaching. Each chunk builds a small tibble, shows the first 10 rows with `head(..., 10)`, then plots all rows.

Strong positive, roughly linear

Context: Saplings in a plantation. Stem diameter (cm at breast height) vs age (years). Older stems are thicker; points fall near a straight line with little scatter.

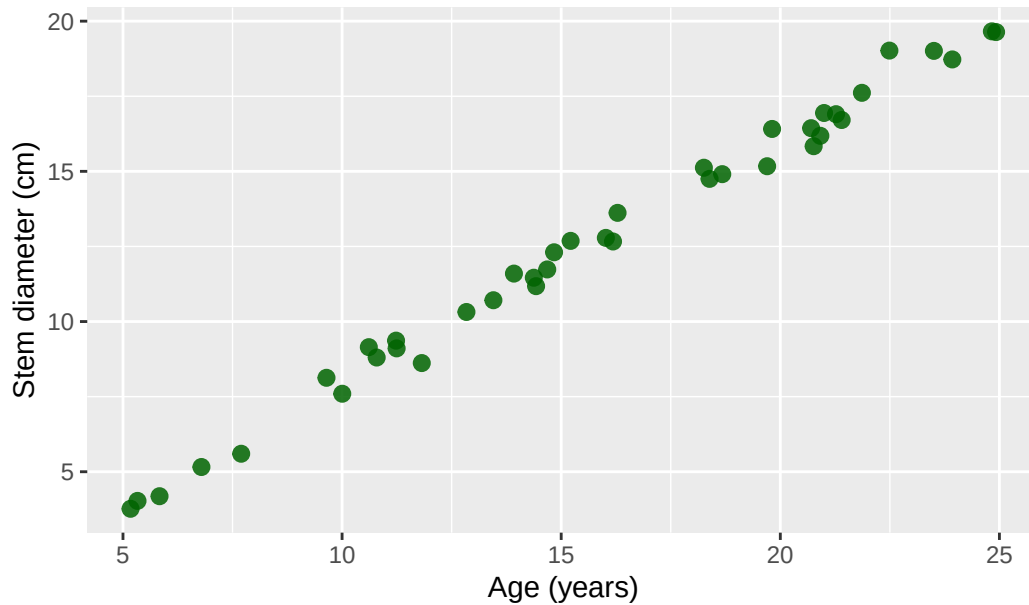
```
set.seed(26)
trees_age_diam <- tibble(
  age_years = runif(40, 5, 25),
  diameter_cm = 0.8 * age_years + rnorm(40, 0, 0.4)
)

head(trees_age_diam, 10)
```

```
# A tibble: 10 x 2
  age_years diameter_cm
    <dbl>         <dbl>
1     5.33         4.03
2    10.8         8.80
3    22.5        19.0
4    21.0        16.9
5    11.2         9.11
6    14.4        11.2
7    20.9        16.2
8    19.7        15.2
9    11.8         8.62
10   18.4        14.8
```

```
ggplot(trees_age_diam, aes(x = age_years, y = diameter_cm)) +
  geom_point(alpha = 0.85, size = 2.5, color = "darkgreen") +
  labs(
    title = "Sapling diameter vs age (illustrative)",
    x = "Age (years)",
    y = "Stem diameter (cm)"
  )
```

Sapling diameter vs age (illustrative)



Words: Positive direction, strong strength, linear form.

Strong negative, roughly linear

Context: Stream sites along a mountain gradient. Elevation (m) vs water temperature (°C) at midday in summer. Higher sites tend to be cooler.

```
set.seed(27)
stream_temp <- tibble(
  elevation_m = runif(35, 200, 2200),
  water_temp_c = 22 - 0.0065 * elevation_m + rnorm(35, 0, 0.35)
)

head(stream_temp, 10)
```

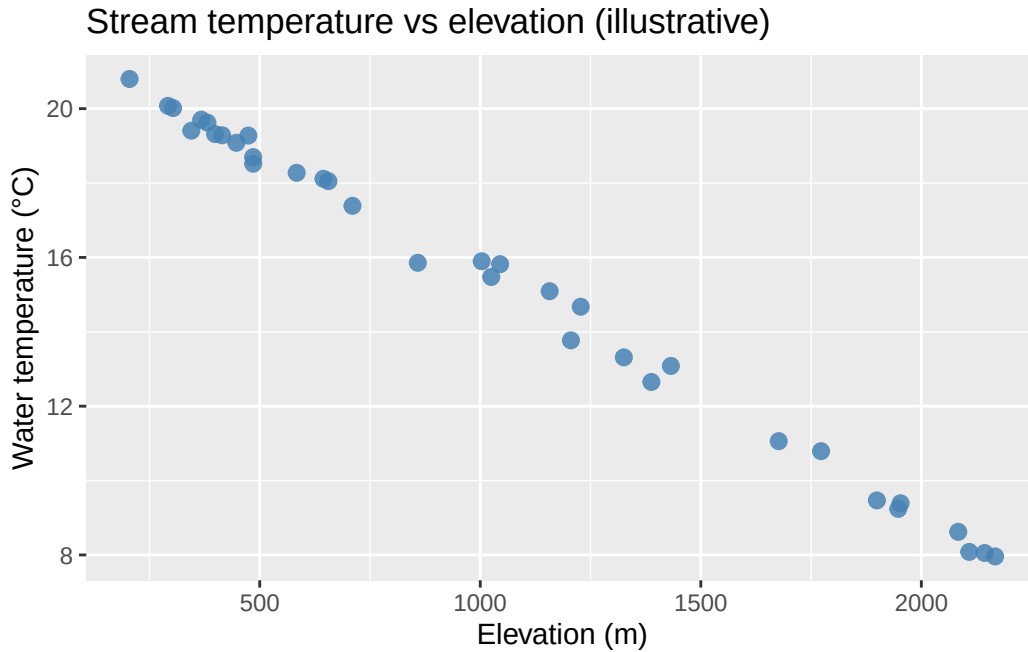
```
# A tibble: 10 x 2
  elevation_m water_temp_c
    <dbl>         <dbl>
1    2144.         8.05
2     368.        19.7
3    1948.         9.24
4     858.        15.9
5     645.        18.1
6    1003.        15.9
7     345.        19.4
8     205.        20.8
9     474.        19.3
10    584.        18.3
```

```
ggplot(stream_temp, aes(x = elevation_m, y = water_temp_c)) +
  geom_point(alpha = 0.85, size = 2.5, color = "steelblue") +
  labs(
```

```

title = "Stream temperature vs elevation (illustrative)",
x = "Elevation (m)",
y = "Water temperature (°C)"
)

```



Words: Negative direction, strong strength, linear form.

Moderate positive, linear

Context: Herbaceous plants along a transect. Soil nitrate (mg/kg) vs leaf nitrogen (%). Higher soil N tends to go with higher leaf N, with noticeable scatter from other factors.

```

set.seed(28)
leaf_soil <- tibble(
  soil_nitrate_mgkg = runif(45, 5, 45),
  leaf_nitrogen_pct = 2.0 + 0.06 * soil_nitrate_mgkg + rnorm(45, 0, 0.35)
)

head(leaf_soil, 10)

```

```

# A tibble: 10 x 2
  soil_nitrate_mgkg leaf_nitrogen_pct
      <dbl>          <dbl>
1         6.14         2.58
2         8.53         2.28
3        24.0         3.50
4        40.4         4.74
5         8.66         2.59
6        35.3         4.07
7         6.38         2.11
8        39.7         4.01
9        27.6         3.67

```

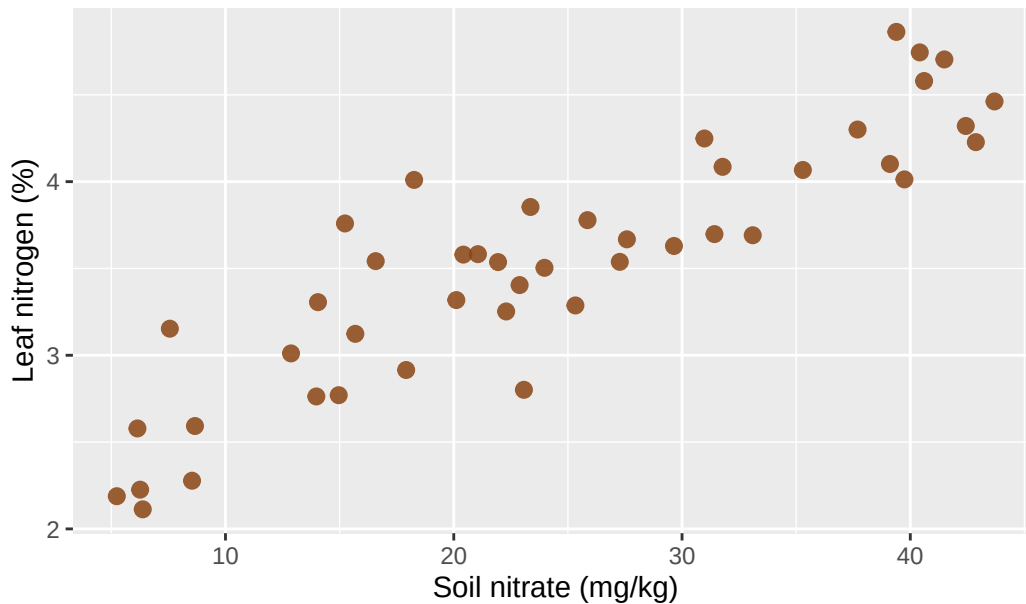
10

42.9

4.23

```
ggplot(leaf_soil, aes(x = soil_nitrate_mgkg, y = leaf_nitrogen_pct)) +
  geom_point(alpha = 0.85, size = 2.5, color = "chocolate4") +
  labs(
    title = "Leaf nitrogen vs soil nitrate (illustrative)",
    x = "Soil nitrate (mg/kg)",
    y = "Leaf nitrogen (%)"
  )
```

Leaf nitrogen vs soil nitrate (illustrative)



Words: Positive direction, moderate strength, linear form.

Weak positive

Context: Forest plots. Canopy openness (percent sky visible) vs understory seedling count per m^2 . Slight tendency for more seedlings where light is higher, but noise dominates.

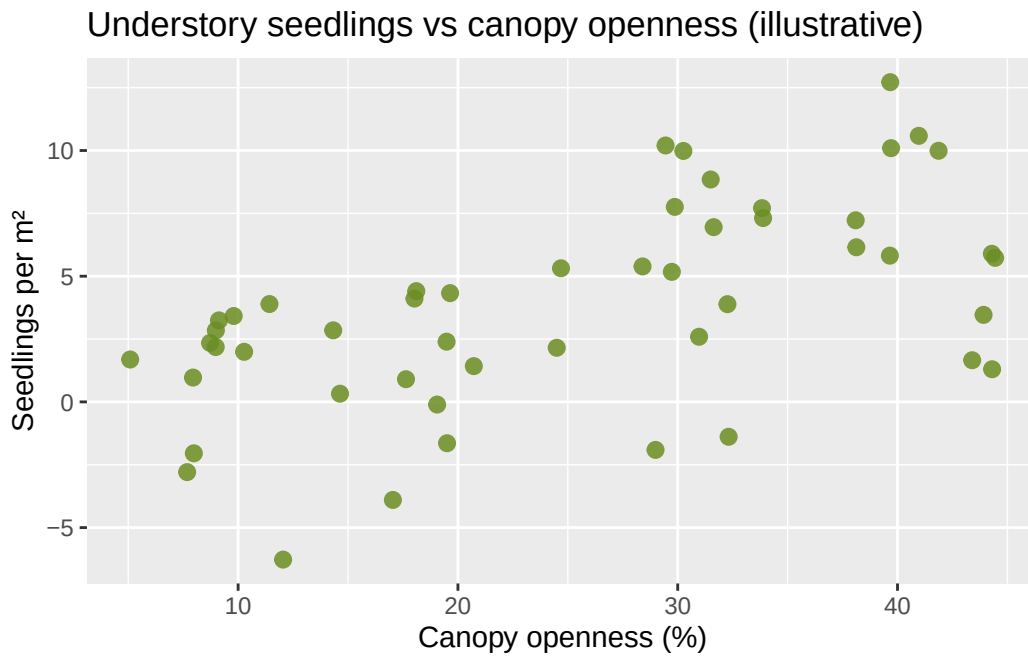
```
set.seed(29)
understory <- tibble(
  canopy_open_pct = runif(50, 5, 45),
  seedlings_per_m2 = 2.5 + 0.08 * canopy_open_pct + rnorm(50, 0, 4)
)

head(understory, 10)
```

```
# A tibble: 10 x 2
  canopy_open_pct seedlings_per_m2
      <dbl>         <dbl>
1      8.99         2.84
2     14.6         0.325
3      9.13         3.24
4     18.0         4.11
5     28.4         5.39
```

6	8.72	2.35
7	38.1	6.15
8	39.7	5.82
9	9.80	3.42
10	14.3	2.85

```
ggplot(understory, aes(x = canopy_open_pct, y = seedlings_per_m2)) +
  geom_point(alpha = 0.85, size = 2.5, color = "olivedrab") +
  labs(
    title = "Understory seedlings vs canopy openness (illustrative)",
    x = "Canopy openness (%)",
    y = "Seedlings per m2"
  )
```



Words: Positive direction, weak strength, linear drift at best.

No clear association

Context: Moose pellet counts along transects vs soil pH. No story linking these two in this toy dataset; points fill the rectangle.

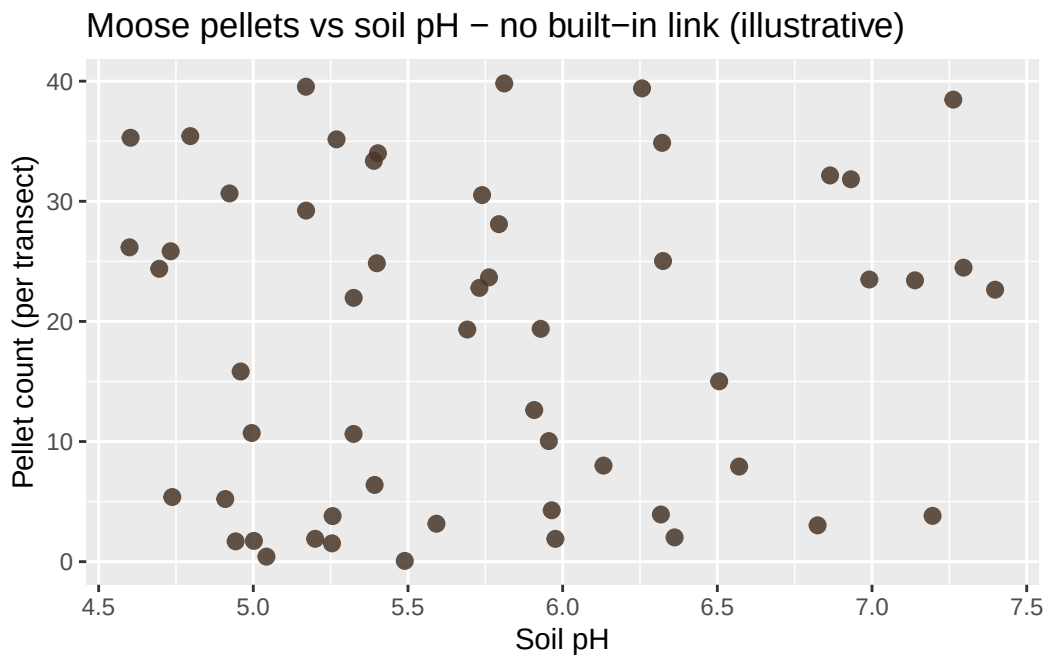
```
set.seed(30)
moose_ph <- tibble(
  soil_ph = runif(55, 4.5, 7.5),
  pellet_count = runif(55, 0, 40)
)

head(moose_ph, 10)
```

```
# A tibble: 10 x 2
  soil_ph pellet_count
  <dbl>      <dbl>
1    4.80         35.4
```

2	5.96	4.28
3	5.59	3.16
4	5.76	23.7
5	5.40	34.0
6	4.94	1.69
7	7.20	3.81
8	5.17	29.2
9	7.40	22.6
10	4.92	30.7

```
ggplot(moose_ph, aes(x = soil_ph, y = pellet_count)) +
  geom_point(alpha = 0.85, size = 2.5, color = "#4a3728") +
  labs(
    title = "Moose pellets vs soil pH - no built-in link (illustrative)",
    x = "Soil pH",
    y = "Pellet count (per transect)"
  )
```



Words: No clear direction, no real strength, no simple form.

Nonlinear (curved): positive, strong

Context: Tree height (m) vs age (years) in even-aged stands. Growth is fast when young and levels off as crowns compete (diminishing returns), so the mean trend bends.

```
set.seed(31)
tree_curve <- tibble(
  age_years = runif(42, 8, 80),
  height_m = 25 * (1 - exp(-0.045 * age_years)) + rnorm(42, 0, 1.1)
)

head(tree_curve, 10)
```

```
# A tibble: 10 x 2
  age_years height_m
  <dbl>     <dbl>
1     45.6     21.6
2     76.4     24.9
3     38.7     21.5
4     38.6     20.5
5     76.0     24.3
6     40.6     23.5
7     68.0     23.8
8     20.4     14.8
9     75.2     24.1
10    25.5     15.6
```

```
ggplot(tree_curve, aes(x = age_years, y = height_m)) +
  geom_point(alpha = 0.85, size = 2.5, color = "forestgreen") +
  labs(
    title = "Tree height vs age - saturating growth (illustrative)",
    x = "Age (years)",
    y = "Height (m)"
  )
```



Words: Overall positive association, strong trend, nonlinear (curved) form.

Nonlinear: hump-shaped (moderate)

Context: Pollinator visits per flower vs nectar volume (μL). Very little nectar attracts few visitors; moderate nectar is best; very high volume may correlate with different flower types or confounding, producing a hump in this toy example.

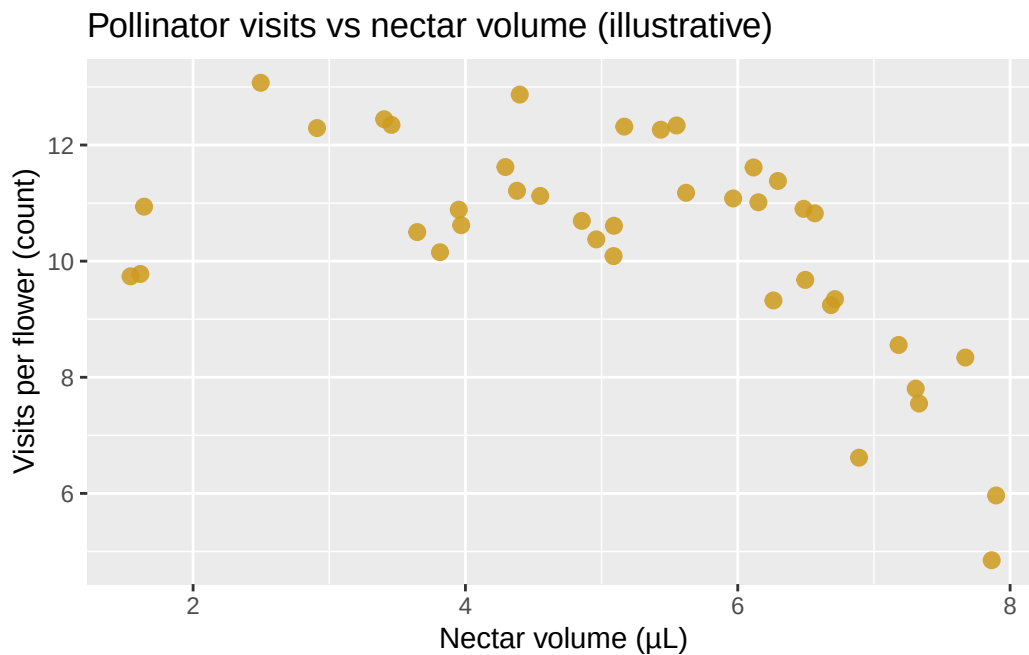
```
set.seed(32)
nectar <- tibble(
```

```
nectar_ul = runif(40, 0.5, 8),
visits = -0.35 * (nectar_ul - 4)^2 + 12 + rnorm(40, 0, 1.2)
)

head(nectar, 10)
```

```
# A tibble: 10 x 2
  nectar_ul visits
    <dbl>   <dbl>
1     4.29  11.6
2     4.96  10.4
3     6.57  10.8
4     5.97  11.1
5     1.64  10.9
6     7.67   8.34
7     6.15  11.0
8     6.89   6.62
9     5.55  12.3
10    3.40  12.4
```

```
ggplot(nectar, aes(x = nectar_ul, y = visits)) +
  geom_point(alpha = 0.85, size = 2.5, color = "goldenrod3") +
  labs(
    title = "Pollinator visits vs nectar volume (illustrative)",
    x = "Nectar volume (µL)",
    y = "Visits per flower (count)"
  )
```



Words: Nonlinear form (hump), moderate strength; direction is not simply positive or negative over the whole x range.

Strong negative, nonlinear

Context: Lake depth (m) vs submerged macrophyte cover (%). In deeper water, light drops; cover falls off quickly at first, then more slowly (curved decline).

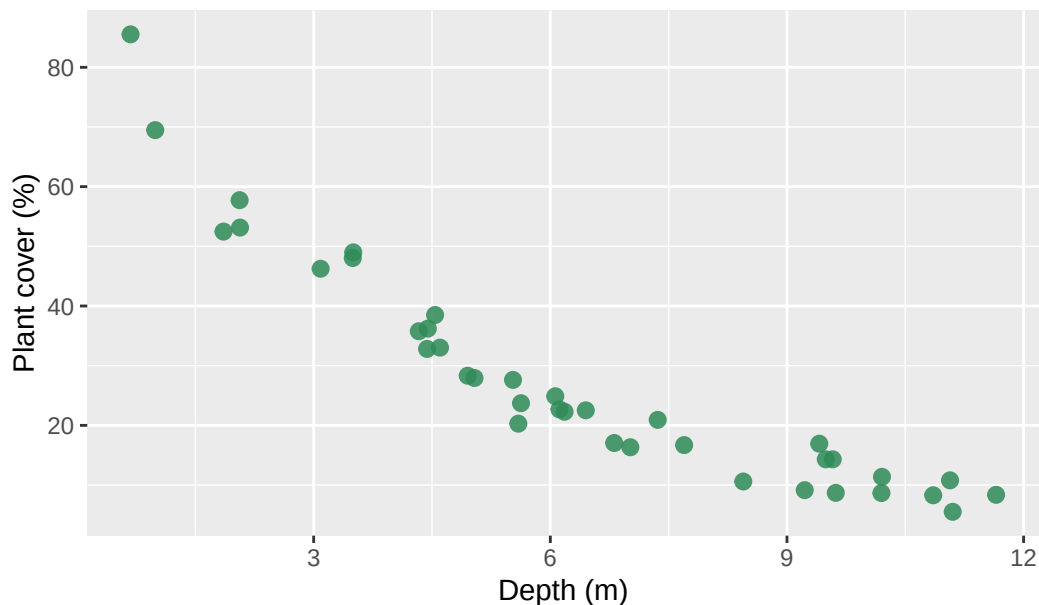
```
set.seed(33)
lake <- tibble(
  depth_m = runif(38, 0.5, 12),
  plant_cover_pct = 90 * exp(-0.22 * depth_m) + rnorm(38, 0, 4)
)

head(lake, 10)
```

```
# A tibble: 10 x 2
  depth_m plant_cover_pct
    <dbl>         <dbl>
1   5.63          23.7
2   5.04          27.9
3   6.06          24.9
4  11.1           10.8
5  10.2           11.4
6   6.45          22.5
7   5.53          27.6
8   4.45          36.2
9   0.678         85.5
10  1.86          52.5
```

```
ggplot(lake, aes(x = depth_m, y = plant_cover_pct)) +
  geom_point(alpha = 0.85, size = 2.5, color = "seagreen") +
  labs(
    title = "Aquatic plant cover vs depth (illustrative)",
    x = "Depth (m)",
    y = "Plant cover (%)"
  )
```

Aquatic plant cover vs depth (illustrative)



Words: Negative association overall, strong trend, nonlinear form.

Summary

For two numerical variables, a scatterplot in ggplot2 uses `aes(x, y)` and `geom_point()`. Describe what you see with form (linear vs curved vs none), direction (positive, negative, or none), and strength (strong through none).

Pearson correlation r (linear association)

When a scatterplot looks roughly linear, add a numerical summary of linear direction and strength: Pearson's correlation coefficient r .

Fact	Detail
Range	$-1 \leq r \leq 1$ (unitless)
Sign	$r > 0$ positive direction; $r < 0$ negative
Strength	$ r $ near 1 = strong linear trend; near 0 = weak or none
When to use	Form is approximately linear — not for strong curves or outliers

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

The numerator tracks whether x and y move together above/below their means; the denominator rescales so r stays between -1 and 1 .

⚠ Correlation describes linear association only

If the scatterplot is curved, r can be misleading (see Example 4 below). Always read the plot first; use r only when form is roughly linear.

💡 Interpret the output — code comes in the demo lab

The chunks below use tidyverse to produce tables and figures. Focus on reading counts, proportions, summaries, and graphs. You will learn how to write these steps in [Demo 1](#) and [Demo 2](#).

Example 1 — Strong positive linear (study hours and exam score)

```
library(tidyverse)
set.seed(3401)
exam_dat <- tibble(
  study_hours = runif(120, 1, 15),
  exam_score = 45 + 3.4 * study_hours + rnorm(120, 0, 8)
)
cor(exam_dat$study_hours, exam_dat$exam_score)
```

```
[1] 0.8441325
```

```
ggplot(exam_dat, aes(x = study_hours, y = exam_score)) +
  geom_point(alpha = 0.75, color = "steelblue4") +
  labs(title = "Exam score vs study hours", x = "Study hours", y = "Exam score")
```

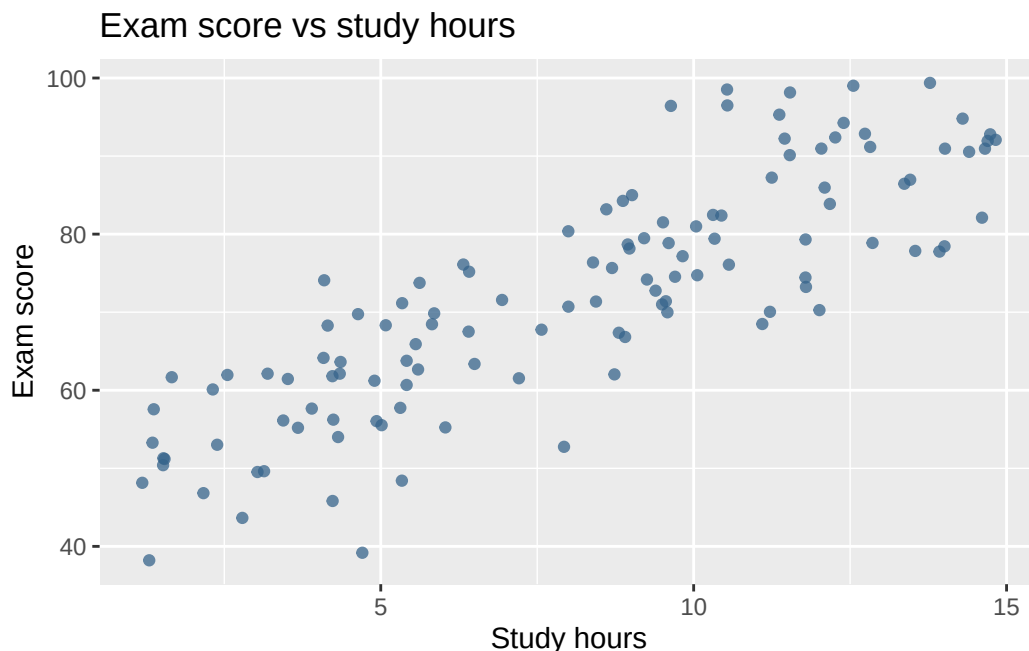


Figure 5: Strong positive linear pattern — points hug an upward line.

Interpret: Form, direction, strength? and does $r \approx 0.9$ match a strong positive linear story?

Answer: Yes: the points hug an upward line (form linear, direction positive, strength strong), so $r \approx 0.9$ matches the picture.

Example 2 — Strong negative linear (elevation and stream temperature)

```
set.seed(3402)
stream <- tibble(
  elevation_m = runif(50, 200, 2200),
  water_temp_c = 22 - 0.0065 * elevation_m + rnorm(50, 0, 0.35)
)
cor(stream$elevation_m, stream$water_temp_c)
```

```
[1] -0.995566
```

```
ggplot(stream, aes(x = elevation_m, y = water_temp_c)) +
  geom_point(alpha = 0.85, color = "steelblue") +
  labs(title = "Stream temperature vs elevation", x = "Elevation (m)", y = "Water temperature (°C)")
```

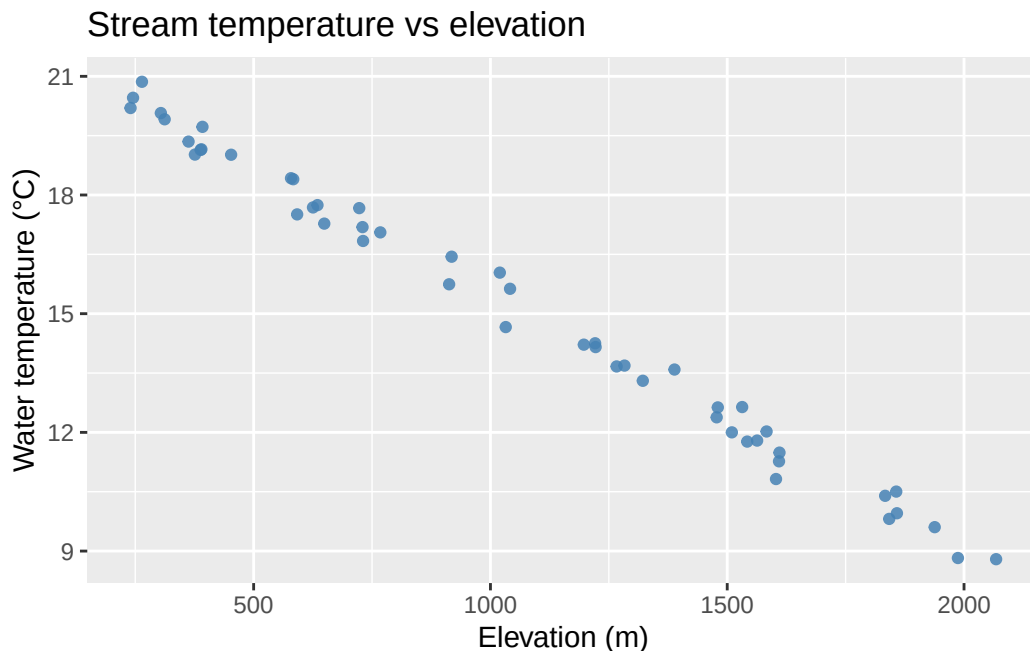


Figure 6: Higher elevation → cooler water; strong negative linear trend.

Interpret: Expect r negative and $|r|$ large. Sign confirms direction; magnitude confirms strength.

Example 3 — Weak positive linear (noisy field data)

```
set.seed(3403)
field <- tibble(
  canopy_open_pct = runif(55, 5, 45),
  seedlings_per_m2 = 2.5 + 0.08 * canopy_open_pct + rnorm(55, 0, 4)
)
cor(field$canopy_open_pct, field$seedlings_per_m2)
```

```
[1] 0.23191
```

```
ggplot(field, aes(x = canopy_open_pct, y = seedlings_per_m2)) +
  geom_point(alpha = 0.85, color = "olivedrab") +
  labs(title = "Seedlings vs canopy openness", x = "Canopy openness (%)", y = "Seedlings per m2")
```

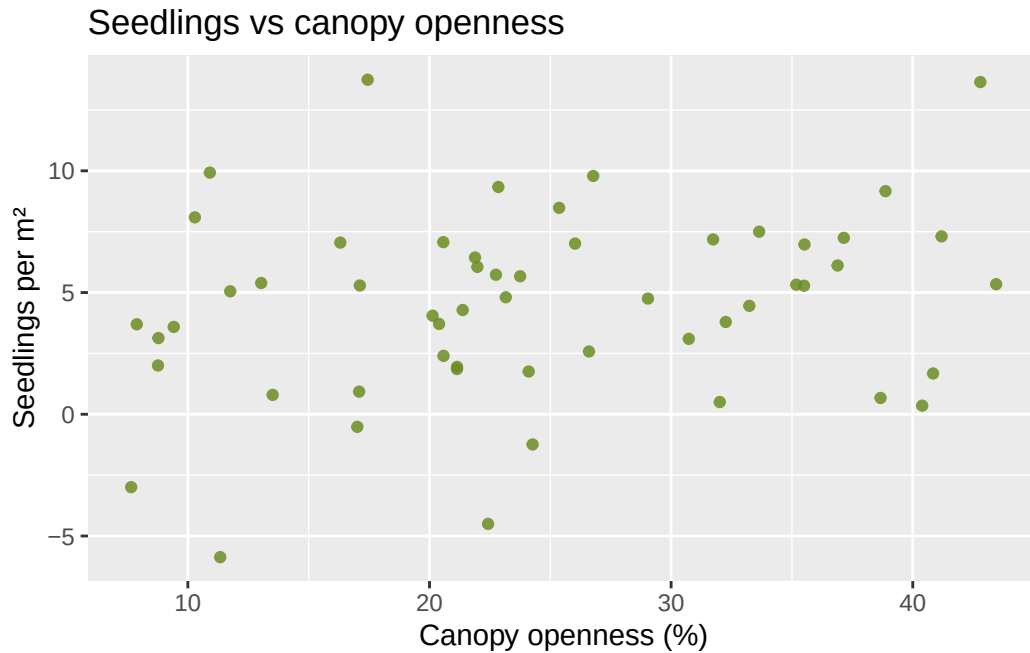


Figure 7: Slight upward drift but wide scatter — weak linear association.

Interpret: r is positive but small in absolute value — matches weak strength despite positive direction.

Example 4 — Curved form: r is not enough (caution)

```
set.seed(3404)
curve_dat <- tibble(
  age_years = runif(45, 8, 80),
  height_m = 25 * (1 - exp(-0.045 * age_years)) + rnorm(45, 0, 1.1)
)
cor(curve_dat$age_years, curve_dat$height_m)
```

```
[1] 0.9167421
```

```
ggplot(curve_dat, aes(x = age_years, y = height_m)) +
  geom_point(alpha = 0.85, color = "forestgreen") +
  labs(title = "Tree height vs age (curved)", x = "Age (years)", y = "Height (m)")
```

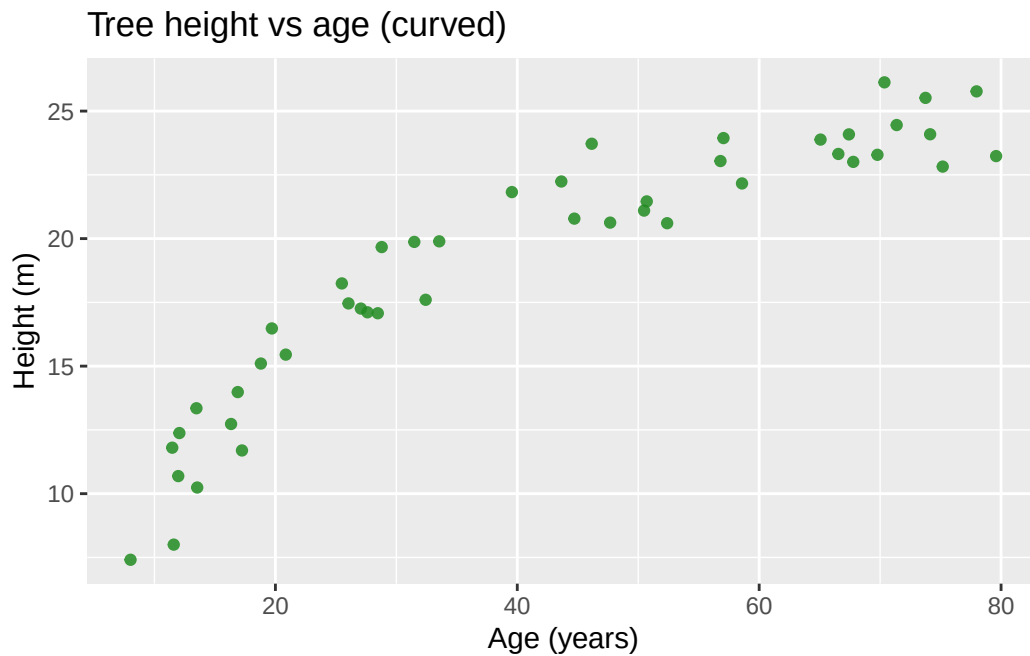


Figure 8: Saturating growth curve — positive association but not a straight-line story.

Interpret: The plot shows nonlinear form. Pearson r may still be positive and moderately large, but it does not fully describe a curved relationship. Report form in words; use r only when linear is reasonable.

i Compute by hand (small datasets)

Work these before checking R. Same numbers appear in [Quiz A](#) and [Synthesis](#) practice banks.

Hand dataset 1 — Hours studied vs exam score ($n = 6$)

Hours x	Score y
1	55
2	62
3	68
4	74
5	80
6	86

Interpret: clear positive linear trend — more hours go with higher scores. Pearson r would be close to $+1$.

Hand dataset 2 — Screen time vs sleep ($n = 6$)

Screen (hr)	Sleep (hr)
1	8.0
2	7.5
3	7.0
4	6.5
5	6.0

Screen (hr)	Sleep (hr)
6	5.5

Interpret: negative linear association — more screen time pairs with less sleep.

Hand dataset 3 — Pearson r from a tiny table ($n = 4$)

x	y
1	2
2	4
3	5
4	8

Points rise mostly linearly (one slight bend at $(3, 5)$). r is positive and moderately strong but not exactly 1. If the scatterplot looks curved, describe form first — r alone is not enough.

i From this sample to a population — inference reference (optional)

Described here: the sample correlation r between two numerical variables. **Population parameter:** ρ (rho) — the true linear correlation in the *whole population*.

If all you have is a random sample but you want to **test whether the variables are linearly related** ($H_0 : \rho = 0$) or **estimate** ρ with a confidence interval, then — after checking conditions (random sample; roughly linear form, no extreme outliers) — use:

```
cor.test(x, y)           # test of rho = 0 and CI for the correlation
```

Introduced later: **Module 4** builds two-group inference, and **Module 5** covers correlation and regression in depth.

Check yourself — Two numerical variables

1. When is a scatterplot appropriate? What goes on each axis?
2. Describe positive direction in a scatterplot.
3. What three things describe association between two numerical variables?
4. Pearson $r = 0.85$ for two variables. What does that suggest?
5. A scatterplot shows a clear curved pattern. Is Pearson r enough?

Answers — Two numerical variables

1. When both variables are numerical. Explanatory on x ; response on y .
2. As x increases, y tends to increase.
3. Form, direction, strength.
4. Strong positive linear association — form should look roughly linear.
5. No — r summarizes linear association only; describe form in words and use r with caution.